# 00 · Foundations — the words the cookbook assumes

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## 1 00 · Foundations — the words the cookbook assumes

The notebook that comes **before** everything else. It builds the core vocabulary — potential outcomes, counterfactuals, confounding, randomization, identification, the do-operator, DAGs, ATE vs CATE — from nothing, with one marketing example, and ends with a worked **estimator ladder**: naive -> regression-adjust -> IPW -> doubly-robust, each recovering a known effect from confounded data.

No new theory here — it mirrors `docs/causal_inference_primer.pdf`. Skim it once and every later notebook reads as a *named tool* rather than new vocabulary.

### 1.1 1 · Treatment, unit, outcome — and the counterfactual

A **treatment** is any action whose effect we want (send a discount, show an ad, change a price). The **unit** receives it (customer, store, region); the **outcome** is what we measure after.

For binary $T \in \{0, 1\}$ each unit has **two potential outcomes**: $Y(1)$ if treated, $Y(0)$ if not. We only observe **one** — $Y = T\,Y(1) + (1 - T)\,Y(0)$ — so the individual effect $Y(1) - Y(0)$ is never seen. This **fundamental problem of causal inference** is a *missing-data* problem; everything downstream is principled imputation of the half of the table we didn't get to see.

| | customer | Y(0) not emailed | Y(1) emailed | individual effect | emailed? | -> OBSERVED | -> MISSING (counterfactual) |
|---|---|---|---|---|---|---|---|
| 0 | Anna | 30 | 80 | 50 | yes | 80 | 30 |
| 1 | Ben | 55 | 60 | 5 | no | 55 | 60 |
| 2 | Cara | 20 | 22 | 2 | yes | 22 | 20 |

### 1.2 2 · From individuals to averages, and the enemy: confounding

Decisions run on **averages**: the **Average Treatment Effect** ATE $= \mathbb{E}[Y(1) - Y(0)]$. The whole difficulty collapses onto one word — **comparable**: to estimate $\mathbb{E}[Y(0)]$ for the treated we need a control group *like them in every way except the treatment.*

**Confounding** breaks comparability. If we email our *loyal* customers and compare to *casual* controls, loyal customers spend more anyway — “loyalty” drove both who got emailed and how much they spend. A **confounder** is a common cause of treatment and outcome; the naive “treated −

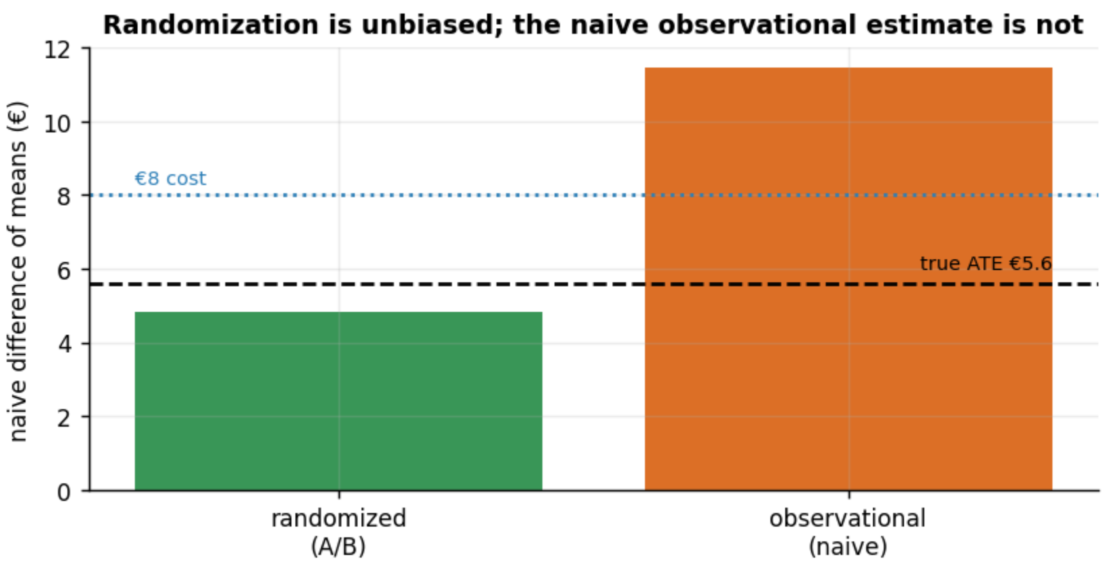
control" difference then blames the treatment for a gap that was really about *who ended up in each group*.

### 1.3 3 · The gold standard, and life without it

**Randomization** (an A/B test) annihilates confounding by design: a coin flip makes treatment independent of every confounder, measured or not, so $\text{ATE} = \langle Y \mid T{=}1\rangle - \langle Y \mid T{=}0\rangle$ — a plain difference of means. We can *see* this: the same simulator run in **randomized** vs **observational** mode gives the same true effect, but only the randomized naive estimate is unbiased.

```
true ATE €5.6  ·  randomized naive €4.8 (unbiased in expectation)  ·
observational naive €11.4 (confounded)

Decision at €8/contact: the TRUE effect €5.6 is BELOW €8, so emailing everyone
does NOT pay.
But the confounded naive €11.4 is ABOVE €8 and would (wrongly) say 'email
everyone' - the same identification error, now flipping a real go/no-go.
```



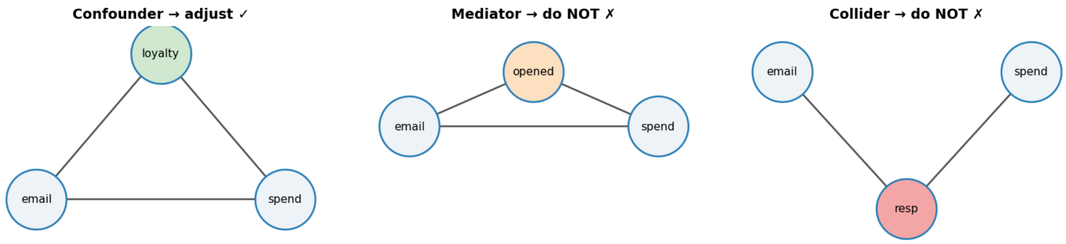

Without randomization we must **identify** the effect using assumptions:

1. **Unconfoundedness** $\{Y(0), Y(1)\} \perp T \mid X$ — within units alike on measured $X$, treatment is as-good-as-random. *(The big, untestable one.)*
2. **Positivity / overlap** $0 < e(x) < 1$, $e(x) = P(T{=}1 \mid X{=}x)$ the **propensity score** — every kind of unit could have gone either way.
3. **SUTVA** — no interference, one version of treatment.

Under these, the **adjustment (g-)formula** compares like-with-like *within* strata of $X$ and averages back up: $\text{ATE} = \mathbb{E}_X[\mathbb{E}[Y \mid T{=}1, X] - \mathbb{E}[Y \mid T{=}0, X]]$.

## 1.4 4 · Seeing vs doing (the do-operator), DAGs, colliders

**Observing** $X = x$ (units where $X$ happened to be $x$, carrying all the reasons it got there) differs from **doing** $do(X = x)$ (reaching in and *setting* it). Causal effects are about *doing.* A **DAG** draws each variable as a node and each direct cause as an arrow, and tells you exactly what to control for:




- **Backdoor path** (arrow *into* T, e.g. $T \leftarrow$ loyalty $\rightarrow Y$): block it by conditioning — the minimal blocking set is the $X$ in the adjustment formula.
- **Mediator** ($T \rightarrow M \rightarrow Y$): controlling for $M$ removes part of the effect you want.
- **Collider** ($T \rightarrow C \leftarrow Y$): conditioning on $C$ *manufactures* a spurious association. Notebook 05 is the whole "what to control for" story.

## 1.5 5 · One effect for everyone, or a different effect for each?

- **ATE** — one number ("did the program lift revenue overall?"): **program evaluation / incrementality** (notebooks 06–11).
- **CATE** $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X{=}x]$ — the effect as a function of *who*: **heterogeneous treatment effects** (notebooks 01–04). If $\tau$ were identical for everyone, there'd be no one to select — targeting works *because* $\tau(x)$ varies.

*(Sometimes a method only recovers the effect for a sub-population — e.g. the compliers an instrument nudges. That restricted quantity is a **LATE**; notebook 11.)*

## 1.6 6 · The estimator ladder — four ways to recover one effect

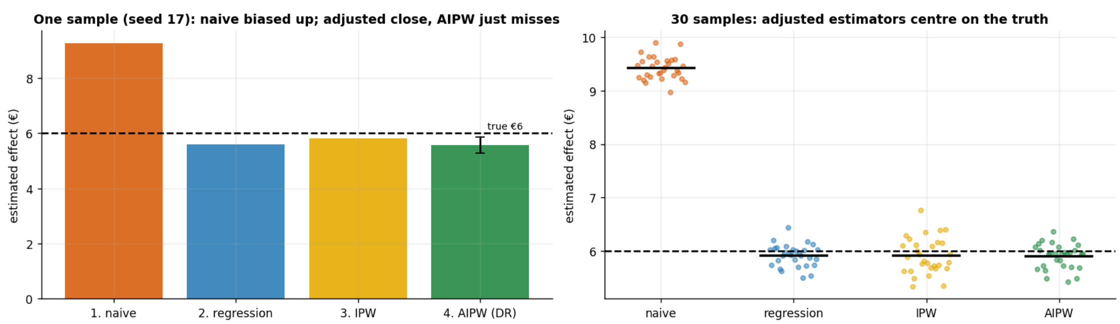
Where the machinery enters. On the **confounded** data (true effect €6), we climb a ladder of estimators, each leaning on different assumptions. On a **single** sample any estimate can land a little off — an honest 90% interval misses about 1 time in 10 — so the test of a *good* estimator is not "exact on this draw" but **unbiased**: centred on the truth *over repeated samples*, while the naive difference stays stubbornly biased. The right-hand panel below shows exactly that across many fresh samples.

1. **Naive** difference of means — ignores $X$, biased up.
2. **Regression adjustment** — model $\mathbb{E}[Y \mid T, X]$, average the contrast (right if the outcome model is right).
3. **IPW** — reweight by the inverse propensity so treated/control look alike (right if the propensity model is right).

4. **AIPW (doubly-robust)** — combine both; consistent if *either* model is right, reported here with a **bootstrap confidence interval** (AIPW is a frequentist estimator — this is *not* a Bayesian credible interval).

```
1. naive          € 9.26   (true €6.0)
2. regression     € 5.60   (true €6.0)
3. IPW            € 5.80   (true €6.0)
4. AIPW (DR)      € 5.57   (true €6.0)

Across 30 fresh samples - mean estimate and bias vs the true €6:
  naive         mean €9.42  bias +3.42  sd 0.21
  regression    mean €5.91  bias -0.09  sd 0.20
  IPW           mean €5.91  bias -0.09  sd 0.33
  AIPW          mean €5.90  bias -0.10  sd 0.23
  AIPW 90% interval covers the truth in 26/30 samples (87%); naive is the only
estimator whose bias does not vanish.
```




The discipline: **Bayes/ML sharpens estimation, never identification.** A tight, beautiful posterior computed under a false unconfoundedness assumption is *confidently wrong*. Priors don't fix broken causal arguments — the graph does. Every notebook keeps identification (this ladder's assumptions) separate from estimation (which rung you climb).

### 1.7 6b · The same lesson on real, famous data (LaLonde 1986)

The ladder above ran on our simulator. Here is the identical confounding story on the most famous real dataset in causal inference — **you don't provide anything; it's fetched from a public URL**:

> **LaLonde / NSW** — a 1970s job-training program. Because it included a **randomized** arm, we know the *true* effect of training on 1978 earnings: about **+$1,800**. But if we throw away the randomized controls and instead compare the trained workers to a *non-experimental* comparison group (people pulled from national surveys), a naive difference is **badly negative** — it looks like training *hurt* earnings, because the trained group was far more disadvantaged to begin with (the confounding). Adjusting for the measured covariates pulls the estimate back toward the true +$1,800. This single dataset is why the field takes confounding seriously.

Gated on `CMP_REAL=1` (keeps the offline test suite deterministic).

```
Real-data section skipped. Set CMP_REAL=1 and re-run to fetch the LaLonde
dataset.
```

### 1.7.1 The eight words to walk in with

**Treatment** · **counterfactual / potential outcome** (the branch you didn't observe) · **confounder** (common cause of treatment and outcome) · **randomization** (kills confounding by design) · **identification** (the assumptions that make an effect recoverable) · **do-operator** (forcing vs observing) · **backdoor / collider** (what to control for, and what never to) · **ATE vs CATE** (one effect for all vs a different effect per customer).

**Next:** 01 · uplift targeting (Anchor A, HTE) and 07 · geo-lift synthetic control (Anchor B, program evaluation) — then the rest of the cookbook.